SER216 Software Engineering Practical Project
Digital Parking Permit System

Madinakhon Tojikarimova
May 4th, 2026

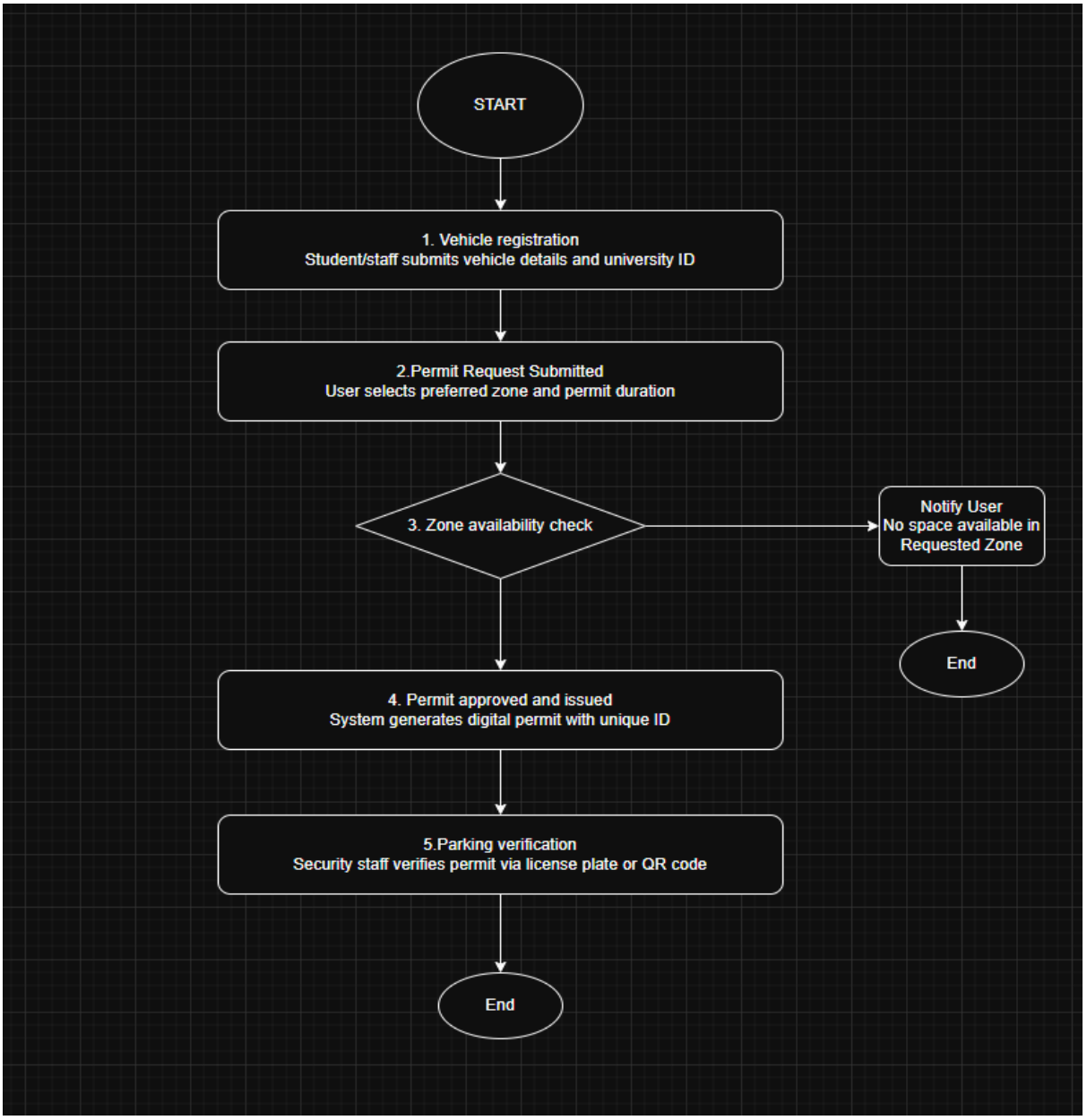
Task 1. Requirements

1.1 *Functional requirements*
1. Vehicle Registration. The system allows students and staff to register one or more vehicles by providing vehicle details (make, model, colour, and licence plate number) linked to their university ID.
2. Parking Permit Request. Registered users shall be able to submit a parking permit application, selecting a preferred parking zone and their desired permit duration (for example, daily, semester, or annual).
3. Zone Availability Check. The system displays real-time availability of all parking zones before a permit is approved, preventing over-allocation of spaces in any zone.
4. Violation Record. Security staff shall be able to search any permit by licence plate number and create, view, or update violation records associated with that permit.

1.2 Non-Functional Requirements
5. Performance The system loads parking zone availability data within 2 seconds under normal network conditions with up to 500 concurrent users.
6. Security All user data and permit records shall be encrypted in transit using TLS 1.2 or higher, and system access shall be controlled by role-based permissions (student / staff / security)

Task 2. Workflow diagram

The workflow diagram below was designed in diagrams.net and illustrates the end-to-end process of the Digital Parking Permit System, starting from vehicle registration through to security verification. The diagram follows standard flowchart notation: rectangles represent processes and diamonds represent decision points



rectangles = processes
diamonds = decision points

Task 3. Test Case Table

Test ID	Feature	Input / Data	Expected Result	Test Type
TC-01	Vehicle Registration + Permit Flow	Valid student ID; Licence plate: ABC123; Zone A selected	Permit issued and stored in database with unique permit ID	System
TC-02	Zone Capacity Check + Denial	Zone B at full capacity; User requests Zone B permit	System denies permit; Displays 'No Space Available in Zone B'	System
TC-03	End-to-End Student Permit Experience	Student enters all required vehicle details and submits permit request	Permit approved; student receives confirmation and can park in assigned zone	Acceptance
TC-04	Security Permit Verification	Security scans licence plate of vehicle with active permit	System displays 'Valid Permit' with zone, permit ID, and expiry date	Acceptance
TC-05	Vehicle Registration After Bug Fix	Same valid inputs used before a previous registration error	System registers vehicle successfully without errors or duplicates	Regression
TC-06	Permit Status After Expiry Bug Fix	Valid permit previously displayed as 'Expired' due to defect	System correctly displays permit status as 'Active' with correct expiry date	Regression

The six test cases above cover the 3 required testing categories:

- System Testing (verifies the integrated system end-to-end),
- Acceptance Testing (verifies the system meets user and business expectations),
- Regression Testing (verifies that previous defects remain fixed after updates)

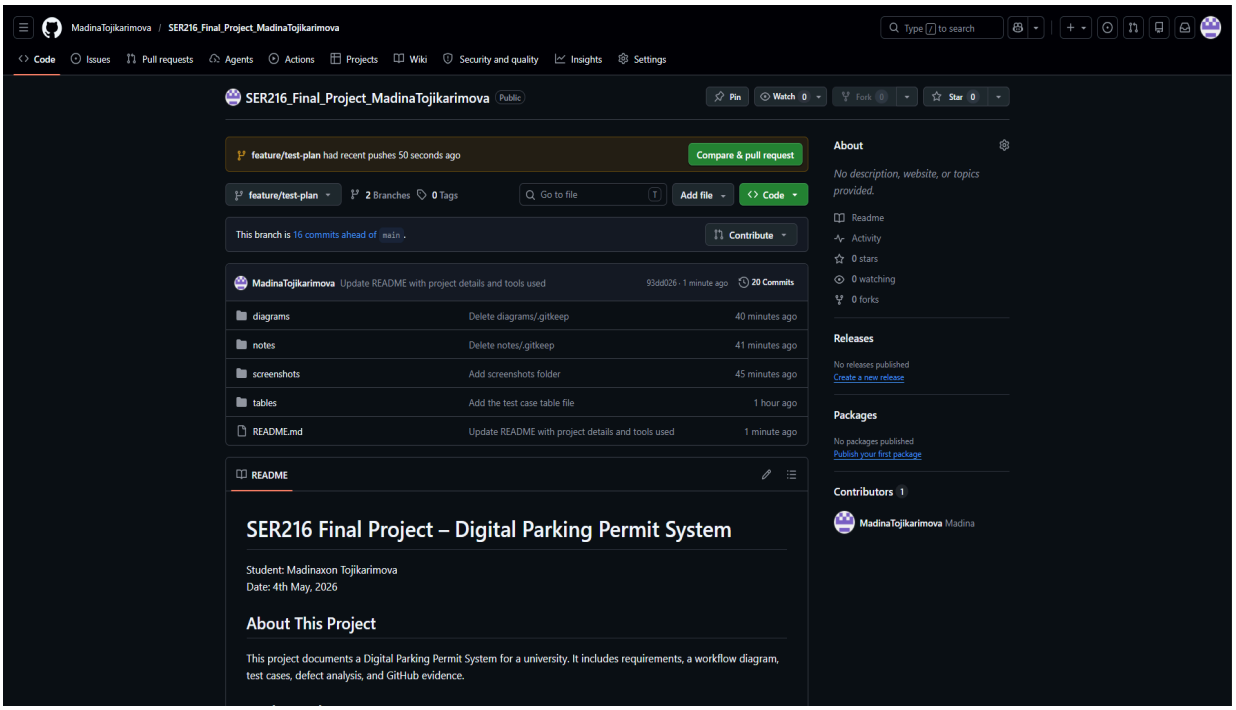
Task 4. Defect Analysis

A valid parking permit is shown as expired during security verification.

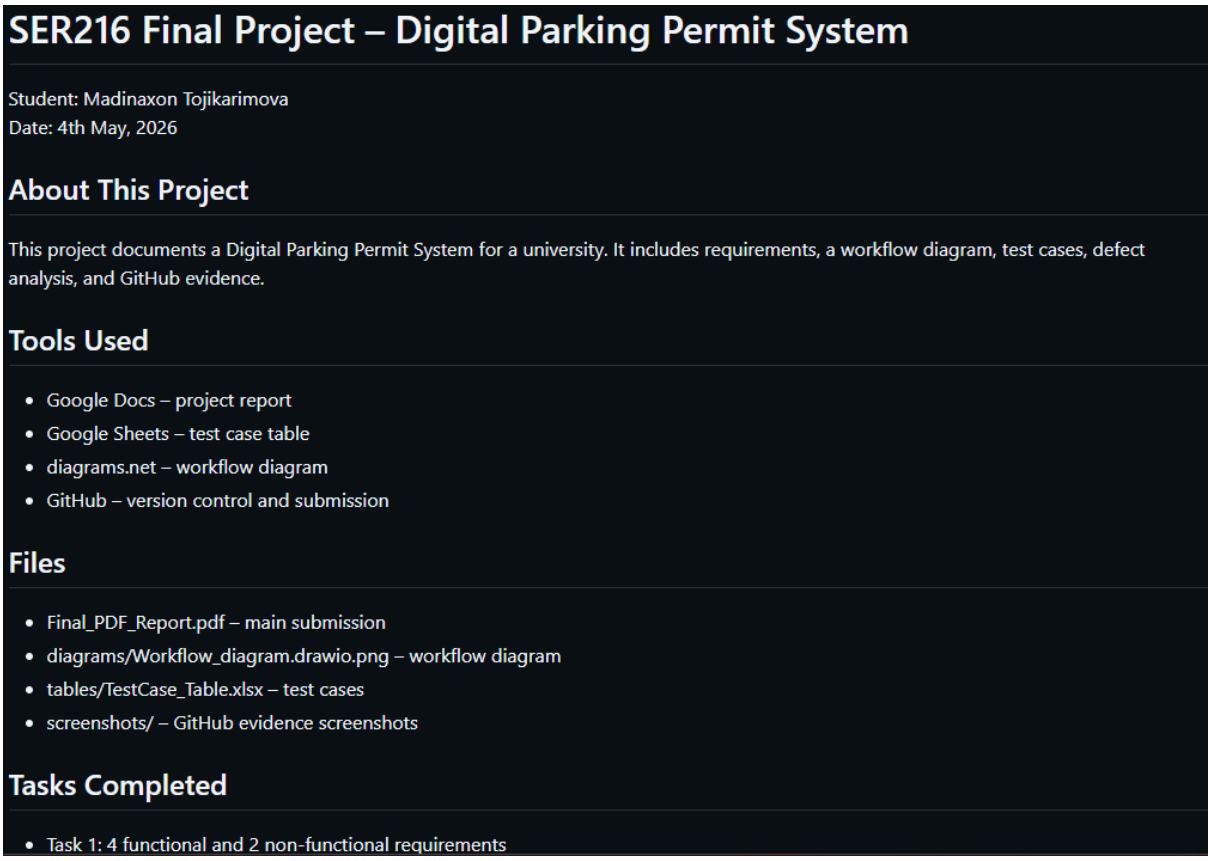
Defect Category	Logic or data error, the system incorrectly evaluates the permit expiry condition.
Severity	High — security staff are unable to verify legitimate permits, which directly blocks normal parking operations and potentially causes incorrect penalty enforcement.
Possible root cause	Incorrect data comparison logic in the verification module. Likely causes include: (1) a timezone mismatch between the database server and the application layer, (2) an off-by-one error in the expiry date calculation, or (3) a date format mismatch causing incorrect parsing
Regression Test	After the fix is deployed, retest permit status display for three boundary cases: (a) a permit expiring today, which must show 'Active', (b) a permit that expired yesterday, which must show 'Expired', and (c) a permit with a future expiry date, which must show 'Active'. These checks must be repeated after every subsequent release to prevent recurrence.

Task 5. Github Evidence

Repo homepage:



Readme.md:



SER216 Final Project – Digital Parking Permit System

Student: Madinaxon Tojikarimova
Date: 4th May, 2026

About This Project

This project documents a Digital Parking Permit System for a university. It includes requirements, a workflow diagram, test cases, defect analysis, and GitHub evidence.

Tools Used

- Google Docs – project report
- Google Sheets – test case table
- diagrams.net – workflow diagram
- GitHub – version control and submission

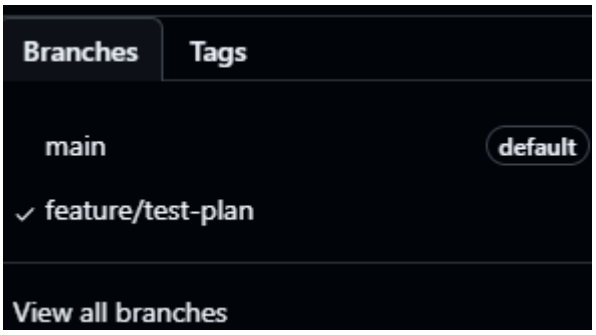
Files

- Final_PDF_Report.pdf – main submission
- diagrams/Workflow_diagram.drawio.png – workflow diagram
- tables/TestCase_Table.xlsx – test cases
- screenshots/ – GitHub evidence screenshots

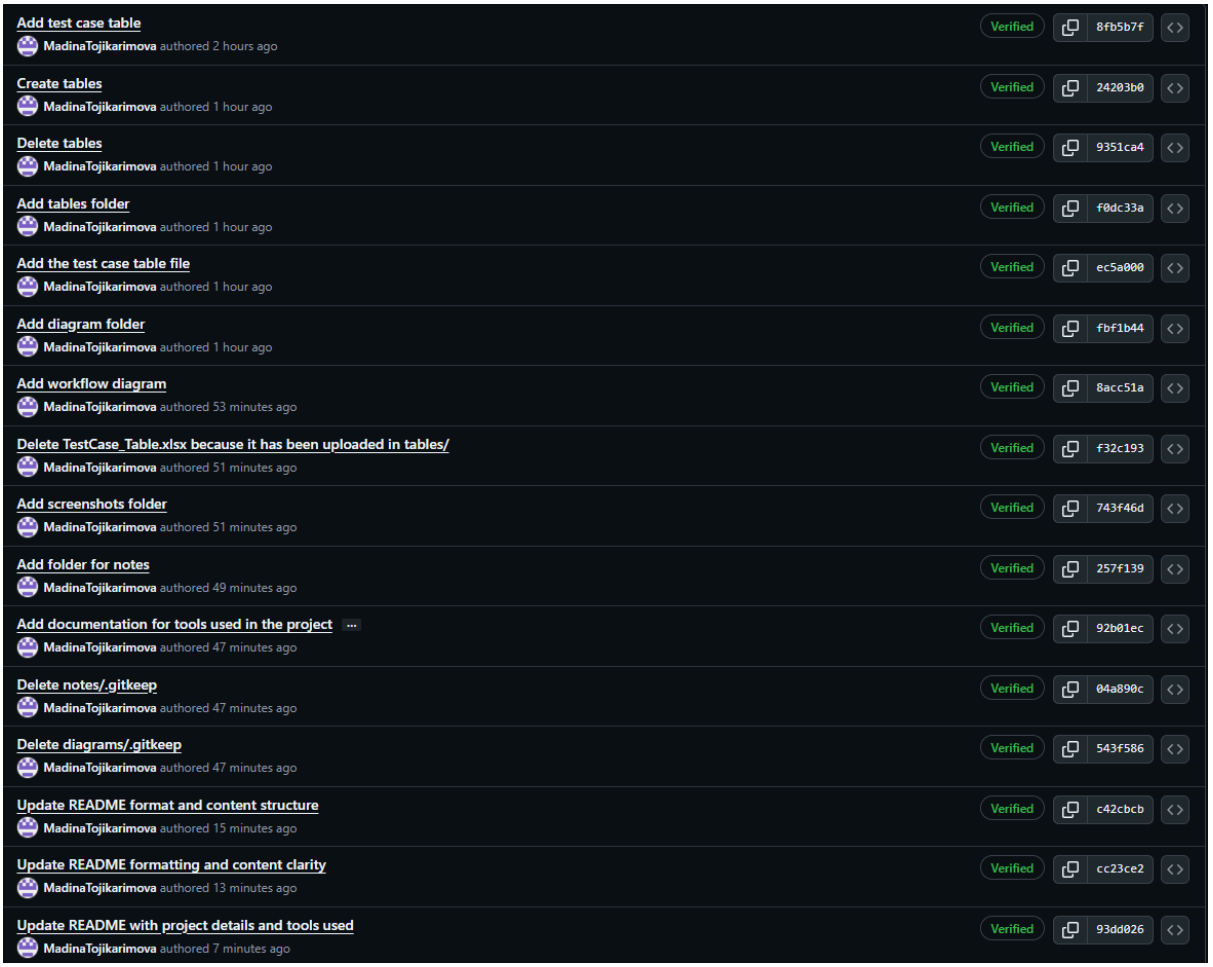
Tasks Completed

- Task 1: 4 functional and 2 non-functional requirements

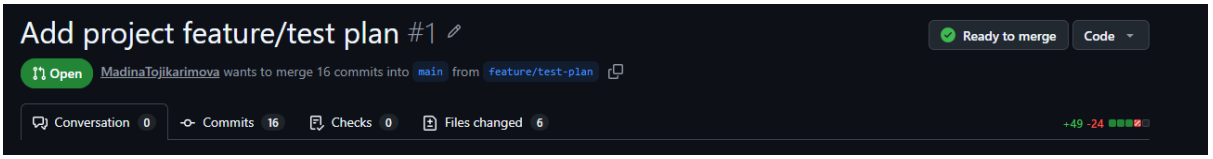
branch:



commits:



pull request:



merged:

https://github.com/MadinaTojikarimova/SER216_Final_Project_MadinaTojikarimova